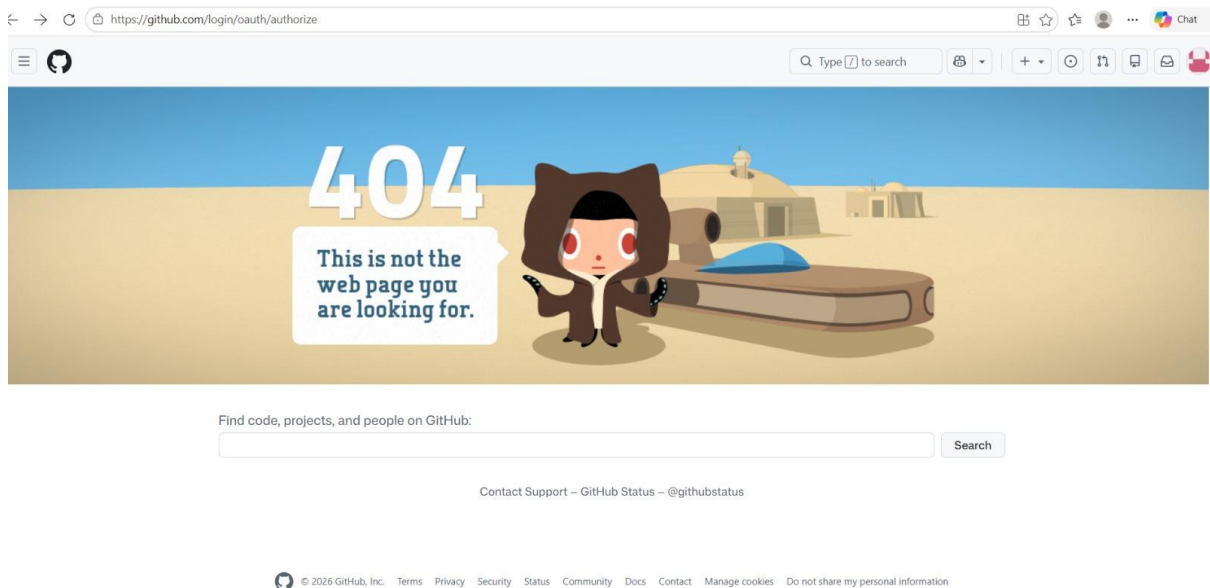


Git Commands and Their Meaning



1. Introduction to Git

What is Git?

Git is a **distributed version control system** used to track changes in files, manage different versions of a project, and collaborate with other developers.

What is GitHub?

GitHub is an online platform that stores Git repositories and allows developers to share and collaborate on projects.

- `git --version`

Purpose:

Checks the installed Git version.

- `git config --global user.name "Your Name"`

Purpose:

Sets the username for Git commits.

```
Run

git config --global user.email "you@example.com"
git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'sampa@Sampada Aher415.(none)')
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git config --global user.name "sampadaaaher415"
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git config --global user.email "sampadaaaher415@gmail.com"
```

- `git config --global user.email "your@email.com"`

Purpose:

Sets the email address associated with Git commits.

```
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git config --global user.email "sampadaaher415@gmail.com"
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git config --global --list
user.name=sampadaaher415
user.email=sampadaaher415@gmail.com
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git commit -m "Initial commit"
[main (root-commit) 7bd109c] Initial commit
4 files changed, 120 insertions(+)
create mode 100644 .gitignore
create mode 100644 index.html
create mode 100644 script.js
create mode 100644 style.css
```

- `git init`

Purpose:

Creates a new Git repository in the current folder.

- `git add .`

Purpose:

`git add` is used to stage changes before creating a commit. It tells Git, "Include these specific changes in my next commit."

```
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git add .
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   .gitignore
        new file:   index.html
        new file:   script.js
        new file:   style.css
```

- `git status`

Purpose:

Shows the current state of files.

```

PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   .gitignore
    new file:   index.html
    new file:   script.js
    new file:   style.css

```

- `git add filename`

Purpose:

Adds a specific file to the staging area.

- `git commit -m "message"`

Purpose:

Saves changes permanently with a description.

```

user.name=sampadaaher415
user.email=sampadaaher415@gmail.com
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git commit -m "Initial commit"
[main (root-commit) 7bd109c] Initial commit
4 files changed, 120 insertions(+)
create mode 100644 .gitignore
create mode 100644 index.html
create mode 100644 script.js
create mode 100644 style.css
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git branch -M main
>> git push -u origin main

```

- `git remote add origin URL`

Purpose:

Connects local repository with a GitHub repository.

```

Please make sure you have the correct access rights
and the repository exists.
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git remote add origin https://github.com/sampadaaher415/fullstack-assign.git
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git remote -v
origin https://github.com/sampadaaher415/fullstack-assign.git (fetch)
origin https://github.com/sampadaaher415/fullstack-assign.git (push)
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git push -u origin main
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git remote add origin https://github.com/sampadaaher415/fullstack-assign.git
error: remote origin already exists.
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git push -u origin main
info: please complete authentication in your browser...

```

- `git remote -v`

Purpose:

Displays connected remote repositories.

```

PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git remote -v
origin https://github.com/sampadaaher415/fullstack-assign.git (fetch)
origin https://github.com/sampadaaher415/fullstack-assign.git (push)
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git push -u origin main
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git remote add origin https://github.com/sampadaaher415/fullstack-assign.git
error: remote origin already exists.
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads

```

- `git remote remove origin`

Purpose:

Removes the connection with a remote repository.

- `git push`

Purpose:

Uploads committed changes to GitHub.

- `git push -u origin main`

Purpose:

First-time push and connects local branch with remote branch.

```

Delta compression using up to 12 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 1.18 KiB | 120.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/sampadaaher415/fullstack-assign.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker>

```

- **Complete command execution**

```

C:\Users\sampa\OneDrive\Desktop> cd Password strength checker
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

..../..../
..../..../vertices

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git rev-parse --show-toplevel
C:/Users/sampa
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> Remove-Item -Recurse -Force C:\Users\sampa\.\git
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git init
Initialized empty Git repository in C:/Users/sampa/OneDrive/Desktop/Password strength checker/.git/
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git rev-parse --show-toplevel
C:/Users/sampa/OneDrive/Desktop/Password strength checker
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        index.html
        script.js
        style.css

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git add .
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   .gitignore
        new file:   index.html
        new file:   script.js
        new file:   style.css

```

```

PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git commit -m "Initial commit"
Author identity unknown

*** Please tell me who you are.

Run

  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'sampa@sampadaa415.(none)')
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git config --global user.name "sampadaa415"
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git config --global user.email "sampadaa415@gmail.com"
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git config --global --list
user.name=sampadaa415
user.email=sampadaa415@gmail.com
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git commit -m "Initial commit"
[main (root-commit) 7bd109c] Initial commit
4 files changed, 120 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 index.html
 create mode 100644 script.js
 create mode 100644 style.css
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git branch -M main
>> git push -u origin main
fatal: 'origin' does not appear to be a git repository
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git remote add origin https://github.com/sampadaa415/fullstack-assign.git
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git remote -v
origin https://github.com/sampadaa415/fullstack-assign.git (fetch)

```

```

user.email=sampadaa415@gmail.com
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git commit -m "Initial commit"
[main (root-commit) 7bd109c] Initial commit
4 files changed, 120 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 index.html
 create mode 100644 script.js
 create mode 100644 style.css
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git branch -M main
>> git push -u origin main
fatal: 'origin' does not appear to be a git repository
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git remote add origin https://github.com/sampadaa415/fullstack-assign.git
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git remote -v
origin https://github.com/sampadaa415/fullstack-assign.git (fetch)
origin https://github.com/sampadaa415/fullstack-assign.git (push)
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git push -u origin main
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git remote add origin https://github.com/sampadaa415/fullstack-assign.git
error: remote origin already exists.
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 1.18 KiB | 120.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/sampadaa415/fullstack-assign.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
PS C:\Users\sampa\OneDrive\Desktop>Password strength checker> █

```


- **Authentaction Succeeded**

If GitHub shows “Authentication succeeded”, it means your GitHub account has been successfully verified and your Git/GitHub login is working.

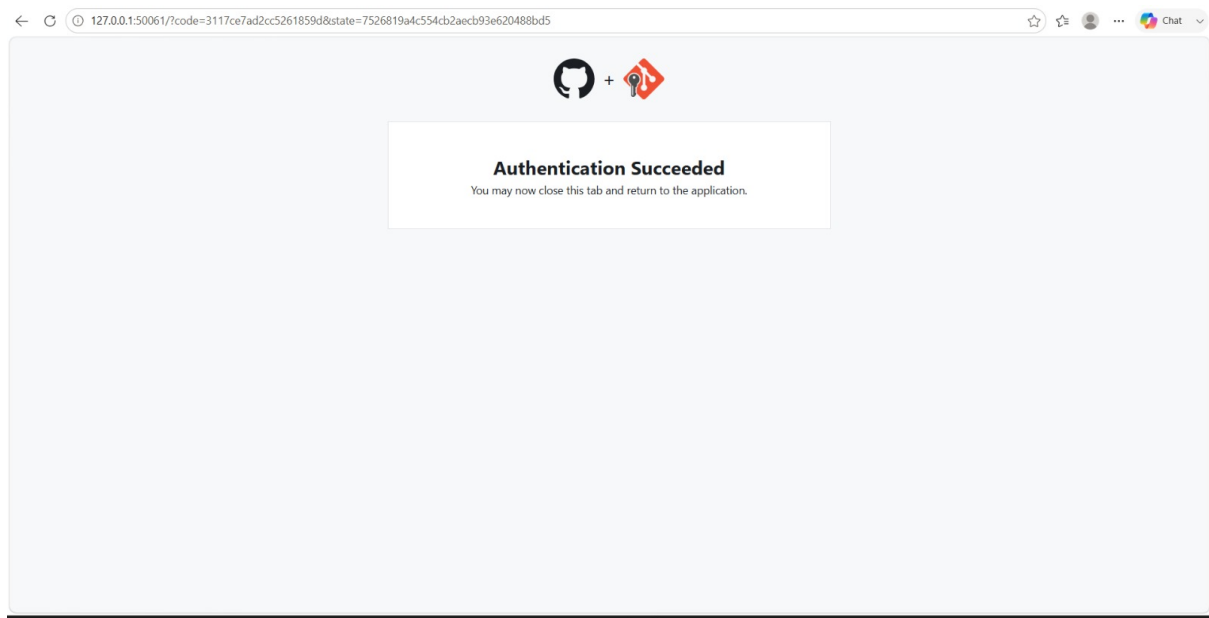
You can now use commands like:

1.git add .

2.git commit -m "first commit"

3.git push

So, Authentication succeeded = GitHub login/verification was successful.



- **Push Repository**

Push means uploading your local project/files from VS Code (your computer) to a GitHub repository.

Use:

1.git add .

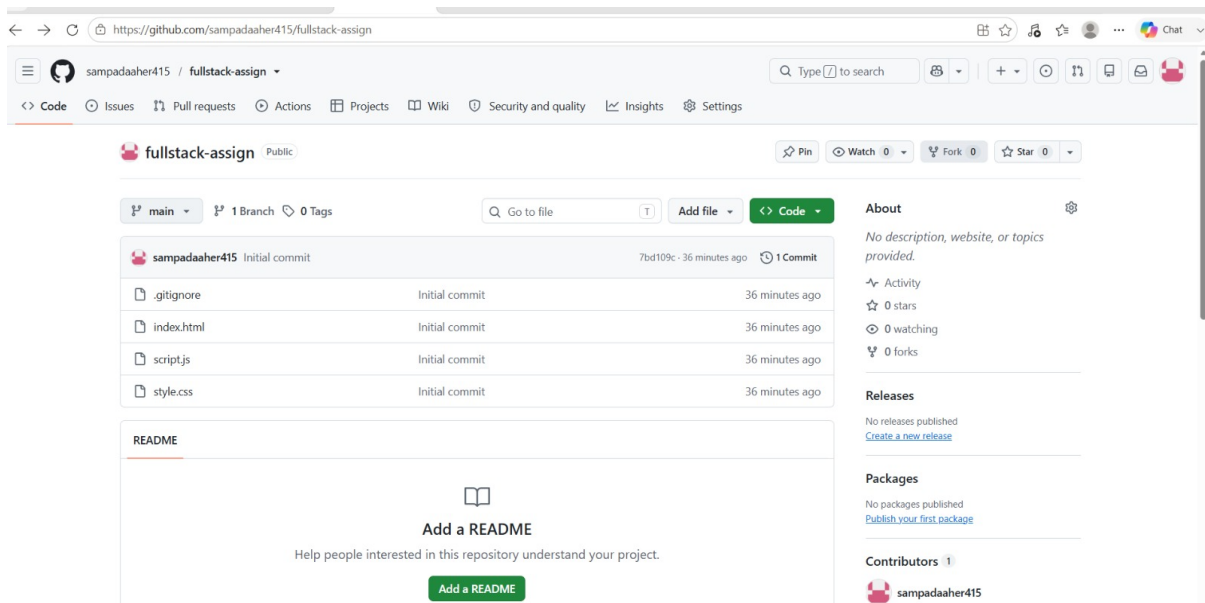
2.git commit -m "Initial commit"

3.git push origin main

Meaning:

- git add . - Selects all changed files.
- git commit -m "Initial commit" - Saves the changes locally.
- git push origin main - Uploads the committed changes to GitHub.

After a successful push, refresh your GitHub repository and you should see your project files there.



- **Conclusion:**

Thus, we **successfully linked Visual Studio Code with GitHub using Git commands**. We configured Git, authenticated our GitHub account, initialized the repository, added and committed the files, and pushed the project to GitHub. Hence, the project was successfully uploaded and connected to the GitHub repository.